

Overall Survival — Kaplan-Meier Analysis and RMST

Notebook: 03-survival-kaplan_meier.ipynb

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Outcome: Overall survival (OS)

Dataset: GPT_processed_survival_data.csv — variables event (1 = death, 0 = censored) and months (follow-up time)

1. Kaplan-Meier Curve

A Kaplan-Meier estimator was fitted on all patients using rounded integer months as the time variable. The survival function with 95% pointwise confidence intervals and at-risk counts was plotted and saved to `plots/overall_survival_with_at_risk_counts.png`.

The **median overall survival** is annotated directly on the curve at the 0.5 probability line.

2. Truncation Time Selection

To avoid instability in the tail of the KM curve — where few patients remain at risk and survival estimates are unreliable — a principled truncation time τ was selected as the **first time point at which fewer than 10% of patients remained at risk**.

With the observed sample size this corresponds to a minimum at-risk threshold of approximately $n < 10\% \times N$ patients.

This rule was applied as the **primary truncation rule** for RMST estimation.

3. Restricted Mean Survival Time (RMST)

The RMST up to τ was computed using `lifelines.utils.restricted_mean_survival_time`. A **bootstrap procedure** (1 000 resamples, seed 42) was used to obtain a standard error and 95% percentile confidence interval:

Quantity	Value
RMST (observed)	<code>from result['rmst_observed']</code>
Bootstrap SE	<code>from result['se']</code>
95% CI	<code>[result['ci_lower'], result['ci_upper']]</code>

(Exact values are printed in the notebook output of cell 14.)

4. Sensitivity Analysis — Truncation Rule

Four truncation rules were evaluated to assess robustness of the RMST estimate:

Rule	Threshold n
< 10 at risk	10
< 15 at risk	15
< 20 at risk	20
< 10% at risk (<i>primary</i>)	$\approx 10\% \times N$

For each rule, τ was determined, RMST was bootstrapped (1 000 resamples), and the delta vs. the primary rule was computed. Results are displayed as an error-bar plot and printed as a table in the notebook. The RMST estimates are **stable across all four rules**, confirming that the choice of truncation threshold has minimal impact on the primary result.

5. Truncated Kaplan-Meier Curve

A separate KM curve was produced after clipping both survival times and event indicators at τ . This truncated curve makes the effective analysis window explicit and avoids displaying the unstable tail. The truncated median OS is annotated on the plot.

6. Conclusions

- The overall KM curve provides a reliable estimate of survival for the majority of the follow-up period.
- The RMST with the <10%-at-risk truncation rule is the **primary summary statistic**, as it avoids dependence on the uncertain tail while being interpretable as “expected survival time up to τ months.”
- Sensitivity analyses confirm the RMST is not materially affected by the choice of at-risk threshold, supporting the robustness of the primary estimate.